SHANKAR NATH PANDEY

Ph.D. Student | Optimization & AI/ML Researcher

Tampa, FL | +1 (656) 204-4405 | shankar38@usf.edu | shankarpandey2025.github.io

Summary

Ph.D. student in Industrial and Management Systems Engineering at the University of South Florida, working at the intersection of Operations Research, Data Science, and AI/ML. Current research focuses on multiobjective and large-scale optimization, mathematical programming, and decomposition algorithms, with applications in intelligent decision-making and cancer radiation treatment planning.

Education

Ph.D., Industrial and Management Systems Engineering ***2025–Present***
University of South Florida | Tampa, FL

M.Sc. Engineering, Mechanical Design and Manufacturing ***2020–2022***
Tribhuvan University, Institute of Engineering, Thapathali Campus | Nepal

B.E., Mechanical Engineering ***2014–2018***
Tribhuvan University, Institute of Engineering, Pashchimanchal Campus | Nepal

Research Experience

Doctoral Student Researcher | University of South Florida ***2025–Present***

Large-Scale & Multiobjective Optimization — Benders Decomposition
- Developing Benders-SOCP and Benders-LP algorithms for mixed-integer maximum multiplicative programs
- Evaluating efficiency and scalability against direct formulations on randomized benchmarks and traveling-salesperson instances.

AI/ML for Cancer Radiation Treatment Optimization
- Developed a scalable Python–MATLAB pipeline to generate large-scale dose-influence matrices and optimize radiation treatment plans for **prostate cancer cases**
- Developing a **Transformer-based AI/ML framework** integrating reinforcement learning and mathematical optimization to reduce computational complexity while preserving tumor coverage and minimizing radiation exposure to healthy tissues

Distributed Game-Theoretical Learning for Multiplicative Programs
- Exploring distributed learning and game-theoretic approaches to multiplicative optimization.

Teaching Experience

Teaching Assistant | University of South Florida ***Present***
Courses: Manufacturing Process; Automation and Robotics. Supporting laboratory instruction, assessments, and student learning.

Professional Experience

Mechanical Engineer | Nepal Electricity Authority ***Nov. 2021–Dec. 2024***
- Contributed to hydropower plant operations, maintenance, and modernization, including engineering design, project execution, and SCADA-based monitoring, control, and automation systems.

Senior Technician | Nepal Airlines Corporation ***Dec. 2018–Dec. 2021***
- Performed technical maintenance of aircraft ground-handling equipment to support reliable airport operations.

Publication

"Simulation of Propeller Runner for Cylindrical Basin of Gravitational Water Vortex Power Plant."

Conference Presentation

Pandey, S. N. "A Benders Decomposition Approach for Solving a Class of Maximum Multiplicative Programs." *INFORMS Annual Meeting, San Francisco, CA, November 2026.*

Technical Skills

Optimization: Multiobjective optimization; mixed-integer programming; LP, QP, MISOCP; Benders decomposition; convex/nonlinear optimization.
AI/ML: PyTorch; Transformer models; representation learning; optimization-guided learning.
Programming & Solvers: Python, MATLAB, NumPy, SciPy, Gurobi, CVXPY, OSQP, SCS.
Scientific Computing: Large sparse matrices; high-performance computing (SLURM/Linux); computational benchmarking.
Engineering Design & CAD: AutoCAD, SolidWorks, CATIA, ANSYS.

Professional Membership & Recognition

Membership: INFORMS — Institute for Operations Research and the Management Sciences.
Recognition: Government scholarships for engineering studies at Tribhuvan University.